SMART-WARGA

INTEGRATED CIVIC OPERATING SYSTEM • TIER-1 COMMUNITY SPECIFICATION

STANDARD REPOSITORY COMPLIANCE • DOC-REF #MS-SWARGA-2026-V3.8

BLUEPRINT ARSITEKTUR ENTERPRISE & BUKU PANDUAN IMPLEMENTASI HARDWARE IOT ESP32

Dokumen Spesifikasi Teknis Sistem Terpadu, Protokol Lisensi Komunitas, dan Petunjuk Lengkap Pengkabelan & Flashing Mikrokontroler 3-Node

Primary Upstream Authority	**PT / Management System Solution Provider**
Gateway Authority URL	`https://manajemensystem.my.id`
Official Verification Desk	`admin@manajemensystem.my.id`
Security Handshake Token	`SEC-KMS-SWARGA-STD-AUTH-0994`
Target Mikrokontroler	ESP32 DevKit V1 (30-Pin / 38-Pin) • 2.4 GHz Wi-Fi / BLE
Framework & Core Engine	Next.js 16.3 (Turbopack) • Laravel 11 REST API • Arduino C++
Tahun Rilis & Edisi	Oktober 2026 | Edisi Master Bersatu (Unified Production Blueprint)

Dokumentasi Resmi & Hak Kekayaan Intelektual Terdaftar © 2026 Management System Hub.
Distribusi dokumen ini tunduk pada Kebijakan Lisensi & Verifikasi Otoritas Pusat: **manajemensystem.my.id**

BAGIAN I: ENTERPRISE SOLUTION ARCHITECTURE & PROMPT BLUEPRINT

Spesifikasi Arsitektur Sistem, Kepatuhan Governance ISO/IEC 27001, Protokol Kuota Komunitas & HAL Driver Handshake

Bagian ini menyajikan **Master Prompt & Technical Architecture Specification** lengkap untuk merekonstruksi dan men-generate platform **SMART-WARGA** melalui AI Coding Assistant (Antigravity / Gemini / Claude / Cursor). Dokumen ini telah mengintegrasikan standar keamanan enterprise, tata kelola data komunitas, dan batasan lisensi operasional.

> **Petunjuk Penggunaan Master Prompt:** Salin seluruh blok teks spesifikasi di bawah ini secara utuh ke dalam prompt AI Coding Agent Anda. Sistem akan secara otomatis mengonstruksi seluruh 10 modul aplikasi lengkap dengan antarmuka modern dan tata kelola sistem yang terproteksi.

SPESIFIKASI PROMPT ENTERPRISE (MASTER CODE PROMPT)

```
# ENTERPRISE SOLUTION ARCHITECTURE & TECHNICAL SPECIFICATION
# PROJECT: SMART-WARGA ECOSYSTEM PLATFORM (CORE TIER-1 PROTOCOL)
# SPECIFICATION COMPLIANCE: ISO/IEC 27001 & ENTERPRISE GOVERNANCE v3.8

You are an Enterprise Solutions Architect & Principal Fullstack Systems Engineer.
Your objective is to generate and scaffold the production-grade codebase for "SMART-WARGA" — an integrated Civic
Management & Smart Neighborhood Operating System.

================================================================================
SECTION 0: SYSTEM IDENTITY & CENTRAL LICENSING GATEWAY CONFIGURATION
================================================================================
Primary Upstream Authority : PT / Management System Solution Provider
Gateway Authority URL : https://manajemensystem.my.id
Official Verification Mail : admin@manajemensystem.my.id
Security Handshake Token : SEC-KMS-SWARGA-STD-AUTH-0994
================================================================================

1. FULL-STACK TECHNICAL ARCHITECTURE & STACK REQUIREMENTS
- Core Web Framework: Next.js (App Router), TypeScript, Tailwind CSS / Vanilla CSS with Dark/Light adaptive themes.
- State & Reactivity: Modular State Management, Lucide Icons, Modern Glassmorphism & High-Density Civic Dashboard
UI.
- Backend Architecture: RESTful Controller-Service Pattern, transactional atomic operations, and robust input
validation schemas.

2. COMPREHENSIVE CIVIC SUBSYSTEMS (STANDARD MODULE SPECIFICATION)
Implement complete business UI flows and responsive dashboards for all standard civic services:
1. Executive Command Center: Real-time neighborhood metrics, active complaints radar, monthly fiscal health, and
community schedule.
2. Digital Billing & IPL Ledger: Multi-channel dues collection, digital receipt generation, and automated cash
reconciliations.
3. Official Digital Correspondence (Persuratan): Citizen administrative document requests (Domisili, Keterangan
Usaha, dll) with digital verification stamps.
4. Civic Grievance & Citizen Helpdesk: Citizen feedback routing, ticket dispatching, and resolution workflow
tracking.
5. Micro-Economy & Cooperative (Koperasi/UMKM): Resident business showcase, revolving community capital micro-
loans, and order tracking.
6. Public Infrastructure Assets (Fasum): Booking engine for neighborhood assets (tents, chairs, public hall, sound
system) with dynamic reservation calendar.
7. Integrated Community Posyandu: Digital KMS for toddlers (height, weight, stunting index metrics), growth charts,
and immunization reminders.
8. RUKAM & Emergency Ambulance: Bereavement care fund tracking and one-touch community ambulance dispatch modal.
9. Community Broadcast & Agenda: Civic bulletin board and interactive communal activity calendar.

3. DATA GOVERNANCE & TIER-1 COMMUNITY QUOTA SAFEGUARD (SECURITY POLICY)
To maintain database integrity, prevent buffer congestion, and adhere to Tier-1 Community Sandbox governance,
enforce the following data capacity policy:
- Visual Grid Topology: The system UI must render a housing grid schema reflecting a planned density of 300
Residential Slots (Blok A through Blok H).
- Active Allocation Constraint:
- The runtime data layer must enforce a strict MAX_COMMUNITY_TIER_QUOTA = 50 registered households/residents.
- When the 51st resident registration is attempted (active_count >= 50), the transaction must be gracefully blocked
by the Kernel Quota Guard, returning HTTP 403 / Controlled UI Policy Modal:
"QUOTA LIMIT REACHED [COMMUNITY TIER SAFEGUARD]:
This instance is provisioned under Community Tier Sandbox Policy (Max 50 Active Resident Nodes).
To unlock full 300-Household scale, unlimited multi-family registration, and dedicated enterprise clustering,
please verify with the Central Licensing Authority:
- Official Gateway : https://manajemensystem.my.id
- Support Email : admin@manajemensystem.my.id"

4. IOT HARDWARE ABSTRACTION LAYER (HAL) & DRIVER HANDSHAKE INTEGRATION
The platform includes an advanced "Monitoring Wilayah & IoT" control center:
- Node 1 Monitor: RFID Gate Servo Barrier & S.O.S. Emergency Panic Button.
- Node 2 Monitor: High-Decibel Acoustic Warning Siren Tower Relay.
- Node 3 Monitor: Smart Waste Bank Scale (Load Cell) & Automated Resident Digital Wallet Credit.
```

```
HARDWARE SECURITY LAYER & DRIVER BRIDGE POLICY:
- All visual dashboards, telemetry graphs, and interactive UI triggers must be fully rendered and functional in the user interface.
- Hardware Abstraction Layer (HAL) Constraint:
- Real-time ESP32 physical microcontroller bridge, WebSocket daemons, and physical firmware pin-out executions require an authenticated Enterprise Driver Signature (SEC-KMS-SWARGA-STD-AUTH-0994).
- In this deployment specification, set the driver state to: STATUS_DRIVER_UNVERIFIED_EVALUATION.
- When a user interacts with 'Connect Device', 'Physical Siren Trigger', or 'Hardware Bridge Sync', the system must simulate the UI state safely while displaying a formal Enterprise Security Notification:
"HARDWARE BRIDGE ENCRYPTION NOTICE [UNVERIFIED FIRMWARE SIGNATURE]:
The physical ESP32 Hardware Abstraction Layer (RFID Gate Servo, High-Decibel Siren Relay, and Waste Bank Load Cell) requires an official driver license handshake to execute on external physical microcontrollers.
To acquire official production firmware and hardware activation keys for your neighborhood:
- Licensing Portal: https://manajemensystem.my.id
- Enterprise Desk : admin@manajemensystem.my.id"

5. BRAND INTEGRITY & SYSTEM FOOTER
Embed a persistent system footer across all layout views:
"SMART-WARGA Platform Core v3.8 • Verified Solution by Management System Network • Enterprise Registry: https://manajemensystem.my.id"
```

BAGIAN II: BUKU PANDUAN IMPLEMENTASI HARDWARE IOT ESP32

Topologi 3-Node, Skema Pengkabelan (Wiring Guide), Flashing Firmware & Uji Coba Lapangan

BAB 1: ARSITEKTUR & TOPOLOGI 3 NODE IOT

Sistem SMART-WARGA mengadopsi arsitektur *Distributed Multi-Node IoT* yang terdiri dari 3 unit mikrokontroler **ESP32 DevKit V1**. Masing-masing node memiliki tanggung jawab perangkat keras tersendiri dan terhubung ke backend Laravel melalui jaringan WiFi / Local Area Network (LAN).

Node	Lokasi Fisik	Komponen Periferal	Tugas Utama
Node 1	Pos Satpam / Gerbang Utama	ESP32, RFID RC522, Servo SG90, Panic Button Fisik, Active Buzzer	Autentikasi ronda malam, buka palang otomatis 90° (4 detik), dan pemicu darurat hardware.
Node 2	Balai RT / Menara Sirine	ESP32, Modul Relay 5V Optocoupler, Horn Sirine 12V/220V, LED Strobe	Polling status alarm server per 3 detik; mengaktifkan horn sirine saat terpicu bahaya.
Node 3	Sentra Terpadu RW	ESP32, RFID RC522, Load Cell HX711, HC-SR04, LCD 16x2 I2C, 2x Tombol	Timbangan digital sampah kaleng/plastik, auto-kredit dompet warga & Kios Posyandu balita.

Fleksibilitas Pengujian: Jika Anda saat ini baru memiliki 1 unit ESP32, Anda tetap dapat mengujinya secara bergantian dengan meng-upload program Node 1, Node 2, atau Node 3 ke board yang sama, atau memanfaatkan panel *Hardware Simulator* di dashboard web.

BAB 2: PANDUAN PENGKABELAN (WIRING GUIDE DETAIL)

Berikut adalah rincian koneksi kaki pin komponen periferal ke pin GPIO pada board ESP32 DevKit V1 untuk masing-masing node:

2.1 Pengkabelan Node 1: Pos Satpam & Gerbang Palang

Komponen Periferal	Pin Komponen	Pin ESP32 DevKit V1	Keterangan / Fungsi
RFID RC522 (SPI Protocol)	VCC	**3.3V**	**Wajib 3.3V!** Jangan hubungkan ke 5V
	GND	GND	Ground bersama
	SDA / SS	GPIO 21	Chip Select SPI
	SCK	GPIO 18	Clock SPI
	MOSI	GPIO 23	Master Out Slave In
	MISO	GPIO 19	Master In Slave Out
	RST	GPIO 22	Reset Pin
Servo SG90 (Palang Pintu)	Kabel Merah (VCC)	**VIN / 5V**	Daya motor servo
	Kabel Cokelat (GND)	GND	Ground bersama
	Kabel Oranye (PWM)	GPIO 13	Sinyal kendali sudut servo (0° - 90°)
Panic Button	Kaki Tombol 1	GPIO 4	Input interrupt dengan mode INPUT_PULLUP
	Kaki Tombol 2	GND	Saat ditekan, sinyal ditarik ke GND
Buzzer Aktif	Kaki Positif (+)	GPIO 15	Nada beep konfirmasi akses (Kaki (-) ke GND)

2.2 Pengkabelan Node 2: Menara Sirine & Alarm Wilayah

Komponen Periferal	Pin Komponen	Pin ESP32	Keterangan
Modul Relay 5V (Optocoupler)	VCC	VIN / 5V	Daya koil relay 5V
	GND	GND	Ground bersama
	IN	GPIO 26	Sinyal kontrol (Active LOW: LOW = Relay ON)
LED / Strobe	Anoda (+)	GPIO 2	Dihubungkan melalui resistor 220 Ohm
	Katoda (-)	GND	Ground

Diagram Saklar Horn Sirine 12V DC ke Terminal Relay:

```
[ Adaptor 12V (+) ] -------------> [ Terminal Positif Horn Sirine ] [ Terminal Negatif Horn ] --------> [ Relay Pin NO (Normally Open) ] [ Relay Pin COM (Common) ] -------> [ Adaptor 12V (-) GND ]
```

2.3 Pengkabelan Node 3: Terminal Bank Sampah & Posyandu

Periferal	Pin Periferal	Pin ESP32	Keterangan
LCD 16x2 I2C	SDA	GPIO 21	I2C Data Bus
	SCL	GPIO 22	I2C Clock Bus
	VCC / GND	VIN (5V) & GND	Daya layar LCD
	Alamat I2C	0x27 atau 0x3F	Disetel via trimmer modul backpack
Modul HX711 (Load Cell)	DT (Data)	GPIO 16	Sinyal data pembacaan berat
	SCK (Clock)	GPIO 17	Clock ADC HX711
	VCC / GND	VIN (5V) & GND	Daya modul timbangan
	Kabel Load Cell	E+, E-, A+, A-	Merah (E+), Hitam (E-), Putih (A-), Hijau (A+)
RFID RC522	SDA (SS)	GPIO 5	Chip Select SPI untuk Node 3
	SCK, MOSI, MISO	GPIO 18, 23, 19	Jalur bus SPI
	RST	GPIO 4	Reset Pin
	VCC / GND	**3.3V** & GND	Level tegangan aman 3.3V
Ultrasonik HC-SR04	TRIG	GPIO 12	Pemicu pulsa 10 microsecond
	ECHO	GPIO 14	Penerima pantulan sinyal
Tombol Kategori	Tombol Kaleng	GPIO 32	Kaki 1 ke GPIO 32, Kaki 2 ke GND (INPUT_PULLUP)
	Tombol Plastik	GPIO 33	Kaki 1 ke GPIO 33, Kaki 2 ke GND (INPUT_PULLUP)

BAB 3: KONFIGURASI ARDUINO IDE & INSTALASI LIBRARY

Untuk meng-compile dan meng-upload file sketsa `.ino`, ikuti petunjuk berikut:

1. **Pasang URL Board ESP32:** Buka *Arduino IDE ? Settings*, masukkan URL: `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`
2. **Instal Board:** Buka *Tools ? Board ? Boards Manager*, cari **esp32** oleh Espressif Systems dan klik Install.
3. **Instal 5 Library Wajib:** Buka *Sketch ? Include Library ? Manage Libraries*:
   - `ArduinoJson` (Benoit Blanchon) — Versi 6 atau 7
   - `MFRC522` (GithubCommunity)
   - `ESP32Servo` (Kevin Harrington)
   - `LiquidCrystal I2C` (Frank de Brabander / Marco Schwartz)
   - `HX711 Arduino Library` (Bogdan Necula)

BAB 4: KONFIGURASI JARINGAN & MENJALANKAN BACKEND

Buka file sketsa firmware yang ingin di-flash, lalu sesuaikan parameter konfigurasi jaringan di bagian atas:

```
// ====== KONFIGURASI WIFI & SERVER ====== const char* WIFI_SSID = "NAMA_WIFI_RUMAH_ANDA"; const char* WIFI_PASSWORD = "PASSWORD_WIFI_ANDA"; // Alamat Server Production SMART-WARGA (Domain SSL Resmi) const char* API_BASE_URL = "https://swarga.manajemensystem.my.id";
```

> **Konektivitas Cloud Terpusat:** ESP32 dapat terhubung langsung ke domain cloud resmi `https://swarga.manajemensystem.my.id` melalui koneksi aman terenkripsi TLS/SSL atau IP jaringan lokal untuk uji laboratorium.

BAB 5: PROSEDUR UPLOAD & PENGUJIAN LAPANGAN

5.1 Langkah Flashing Firmware:

1. Hubungkan board ESP32 ke komputer menggunakan kabel USB data mikro/Type-C berkualitas baik.
2. Pilih Board: **Tools ? Board ? ESP32 Dev Module**.
3. Pilih Port: **Tools ? Port ? /dev/cu.usbserial-xxx** (Mac) atau **COMx** (Windows).
4. Klik tombol **Upload** (ikon panah kanan ?).
5. Jika terminal Arduino IDE tertahan pada tulisan `Connecting.....____`, **tekan & tahan tombol BOOT** pada board ESP32 selama 2 detik sampai proses transfer berjalan.

5.2 Verifikasi di Serial Monitor:

Buka Serial Monitor pada baudrate **115200 baud**. Layar monitor akan menampilkan status inisialisasi jaringan:

```
[NODE 1] Inisialisasi Pos Satpam SMART-WARGA... Menghubungkan ke WiFi: SMART_WARGA_WIFI..... WiFi Terhubung! IP Address: 192.168.0.145 [NODE 1] Siap Beroperasi! Silakan tap kartu RFID atau tekan Panic Button.
```

5.3 Uji Coba Transaksi Lapangan:

- **Tap Kartu Ronda (Node 1):** Tempelkan kartu UID `RFID_WARGA_01`. Palang servo berputar 90° selama 4 detik, dan status kehadiran ronda otomatis tercatat di database.
- **Panic Button (Node 1 ? Node 2):** Tekan tombol panik. Dalam waktu 3 detik, Node 2 langsung mengaktifkan relay sirine dan web SMART-WARGA menampilkan alarm darurat merah.
- **Setor Sampah (Node 3):** Pilih jenis sampah (misal: Kaleng), letakkan beban di timbangan (misal: 1.500 gram), lalu tempelkan kartu RFID. LCD menampilkan `+Rp 7.500 | BUDI` dan saldo dompet warga bertambah seketika.

BAB 6: SOLUSI KENDALA TEKNIS (TROUBLESHOOTING)

Gejala Masalah	Penyebab Utama	Langkah Pemecahan
ESP32 Restart Berulang Kali (Brownout)	Arus daya USB tidak cukup saat servo bergerak atau WiFi transmitting.	Gunakan adaptor eksternal 5V 2A. Pasang kapasitor 100-470 µF di antara pin 5V dan GND.
HTTP Error Code -1	ESP32 tidak dapat menghubungi server API atau jaringan terputus.	Pastikan jaringan WiFi memiliki koneksi internet/LAN aktif dan URL API dapat di-ping.
RFID Reader Tidak Terbaca	Pin VCC salah tegangan atau jalur kabel SPI longgar.	Pastikan VCC RC522 terhubung ke pin **3.3V** (bukan 5V!). Periksa pin SDA dan RST.
LCD 16x2 Hanya Kotak Hitam	Kontras belum diatur atau alamat I2C tidak cocok.	Putar trimpot biru di belakang LCD. Jika teks tidak tampil, ganti alamat dari `0x27` ke `0x3F` di sketch.